

Who Bears Disaster Risk? Institutions, Income and the Insurance Protection Gap

What I found

I started this project after noticing, during an internship, how often disaster losses in poorer countries went uninsured. The data backs this up strongly. Across 574 natural disasters in 67 countries between 2000 and 2024, the average uninsured share of losses was 46.7% in high-income countries, 78.9% in upper-middle-income countries, 89.6% in lower-middle-income countries and 96.1% in low-income countries. These are just averages, with no model behind them, and the lower-income figures line up with Swiss Re's estimate that 80–90% of catastrophe losses in emerging economies are uninsured.

What I wanted to know was why, so I tested three explanations: income, financial development (credit-to-GDP) and the strength of legal institutions (rule of law). I added them to the regression one at a time, controlling for year and hazard type throughout.

Income on its own looks important, as a doubling of GDP per capita goes with an uninsured share about 13 percentage points lower. Adding financial development barely changes that, however adding rule of law changes it a lot, and income's effect drops by about two-thirds. In the full model, rule of law is the strongest predictor I have. A one-point improvement on the rule-of-law scale is associated to about 10 percentage points less of the damage going uninsured ($p = 0.005$). Income falls to around 4 points per doubling and is no longer conventionally significant ($p = 0.095$). My reading is that a lot of what looked like an income effect is really an institutions effect. Rich countries tend to have strong courts and enforceable contracts, and insurance is only worth buying if you trust the claim will be paid.

Two results I didn't expect

Bigger disasters are less well covered: doubling a disaster's total damage raises the uninsured share by about 2 points ($p < 0.001$). And the type of hazard matters. Compared with earthquakes, storms are about 18 points better covered ($p = 0.003$) and wildfires about 13 points ($p = 0.077$), while floods aren't clearly different.

How confident I am

The thing that worried me most was missing data. EM-DAT never records an insured loss of zero, so when the field is blank, I can't tell whether nothing was insured or nobody reported it, so I ran the model both ways. Dropping the blanks, income's coefficient was -0.052 . Counting them as zero insurance, which grows the sample to 2,709 disasters, it was -0.018 . The sign stayed the same, but the size shrank by about two-thirds. So I trust the direction of the income result much more than its size.

The institution's result held up better. Swapping the rule of law for government effectiveness gave the same idea, (-0.126 , $p < 0.001$), so it doesn't depend on which governance measure I picked. The variance inflation factors were all under 5, which means my three explanations overlap but the model can still tell them apart. I also re-ran everything as a fractional logit, since my outcome is a share stuck between 0 and 1, and it agreed with the ordinary regression on every sign and significance result. Standard errors are clustered by country throughout.

What I couldn't settle

When I split the sample at median income, income's effect seemed to reverse among richer countries (+0.249). I was excited by this at first, but it isn't significant ($p = 0.121$) and it comes from only 15 countries, which is too few to trust clustered standard errors much. I'm treating it as something worth looking into, not a finding. The time split was more reassuring: the income coefficient was similar before and after 2010 (-0.062 and -0.054), so the result doesn't seem to come from one period.

The bigger problem is measurement. Insured losses are least likely to be reported in the poorest countries, which is exactly where the gap is biggest. That means my main sample may be missing many of the events this project is about. Causation could also run the other way, with growing insurance markets pushing countries to strengthen their institutions. And my explanations for why large disasters are less covered, like policy limits or uninsured public infrastructure, are guesses I can't test with this data.

Everything here is a correlation, not to be confused with causation.

What would settle it

Three things would help most. A natural experiment, like a legal reform that improved contract enforcement in one country but not a similar neighbour, would show whether institutions actually cause better coverage. Better reporting of insured losses in lower-income countries, with real zeros recorded, would fix the missing-data problem. And data that separates public from private assets would show whether big disasters are less covered because they hit infrastructure that's hard to insure.